

UM Museum of Natural History Evolution Exercise

Purpose: Some of the best examples of evolutionary concepts and science communication in action are in natural history museums. Your task is to explore the UM Museum of Natural History (UMMNH) and find examples of evolutionary concepts, observe science communication strategies in action, and gain inspiration for your Evolutionary Concept graphic.

Instructions:

- REVIEW the Evolutionary Concepts List below and select 2 evolutionary concepts to explore for your Evolutionary Concept graphic. There are multiple examples for each topic, so you will have lots of options.
- EXPLORE the museum with your group and use the questions below to help guide you to key exhibits and evolutionary examples within the museum. Your responses to each question do not need to be in complete sentences but should provide a thorough answer. This self-guided tour puzzle will require you to work as a team to do some investigation and problem-solving. **Be sure to look closely at specimens and read the associated information posted in the exhibits.** You may also need to use the internet to help you define terms or look up information.
- TAKE PICTURES of any relevant exhibits that you may want to use for your graphic assignment.
- HAVE FUN & learn something new!

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| 1. What Causes Extinction? | 6. How Do We Move? | 11. Molecular homology |
| 2. Ages of Life | 7. Convergent Evolution | 12. Natural vs. Artificial Selection |
| 3. Law of Succession | 8. Vestigial structures | 13. What Can Teeth Tell Us? |
| 4. Transitional Forms | 9. Sexual dimorphism | 14. Jaws Open New Possibilities |
| 5. Homologous structures | 10. Speciation | |

Bio173 Museum Exercise

1. Who is in your group?

2. What two topics are you exploring in the museum?

Topic 1

Topic 2

3. Define what you think that topic is before you do any detailed reading

Topic 1

Topic 2

4. Explore the Museum. Find several exhibits that embody this evolutionary concept. Choose an exhibit, organism, fossil, story, etc. for each topic that you find interesting, surprising or that taught you something. List the name of the exhibit and explain why you picked it.

Topic 1 Exhibit

Topic 2 Exhibit

5. What communication strategies do you see being employed here? Are there eye-catching graphics, interesting questions, neat specimens, storytelling, timelines, etc. being used to draw you in?

Topic 1 Strategy

Topic 2 Strategy

6. What scientific message are you taking away from this exhibit, i.e., what did it teach you (or what do you think it could teach your audience)

Topic 1 Message

Topic 2 Message

7. Before you leave the Museum, take pictures of both exhibits and share them with your group. This will help you remember the exhibits and choose between them. You do not need to include pictures in your final graphic, but if you do, you must credit the UM Museum of Natural History. When you get back to lab, choose one of your two choices as the final topic for your project. Your GSI may chime in case someone else has the same project idea. Reveal it here.

Final topic

8. Now start planning your project. What background knowledge do you think your target audience has about your topic? What is their familiarity, knowledge, *and misconceptions* related to your concept? This is your starting point
9. What points do you want to make to them? What learning objectives do you have? How will you put it in language that is scientifically accurate but that they will understand? You may need to define some terms along the way. This is the roadmap.
10. Give a couple ideas about graphics you can use in your project. You will have time to work on it in lab in the week following and get feedback from your GSI.